Element A:

The Colonial Middle School recently redesigned their building and with this redesign, eliminated their original aquaponics system. This system was a big help in teaching the students about sustainable agriculture, and without it, the teachers don't have the same resources available to them. The problem at hand is the need for implementing an efficient and cost-effective aquaponics system at Colonial Middle School within a budget of $2000. The current absence of the hydroponics system doesn't allow for students to have the valuable hands-on learning opportunities about sustainable agriculture, and with the implementation of the system, would allow students' minds to wander with ideas of aquaponics.

The problem involves designing and implementing an aquaponics system tailored to the specific needs and constraints of Colonial Middle School. This includes considering factors such as the space available for us to use, the $2000 budget limitation, and what kind of educational objectives they wish to fulfill. It also involves ensuring that the system is not only functional but also educational, providing clear insights into the principles of aquaponics. With specific research into the constraints of the project, we found out that space would not be a major constraint, as we have the entire middle of the school building to construct our project.

There are multiple problems that require careful consideration to ensure its successful implementation. This includes identifying suitable equipment, selecting appropriate fish and plant species, and designing the layout for optimal space utilization. Each of these components contribute to the overall effectiveness and sustainability of the aquaponics system.

The implementation of an aquaponic system addresses the concerns of multiple primary stakeholder groups at Colonial Middle School. For students, it provides hands-on learning opportunities in STEM subjects, particularly biology and environmental science. Teachers benefit from an innovative teaching tool that aligns with educational standards and caters to student engagement. Additionally, the administration gains a sustainable solution that enhances the school's reputation and potentially reduces operating costs.

https://www.theaquaponicsource.com/school-aquaponics

https://aquasprouts.com/blogs/green-inspirations/aquaponics-hydroponics-and-schools-why-it-helps-with-a-hands-on-approach-to-learning

https://www.urbangreenfarms.com.au/post/aquaponics-and-schools-why-it-helps-with-a-hands-on-approach-to-learning

The justification for implementing an aquaponic system is supported by a diverse range of credible sources, including academic research, educational publications, and case studies from similar educational institutions. These sources show the benefits of hands-on learning, the importance of sustainability education, and the effectiveness of aquaponic systems in educational settings. Additionally, the timely relevance of sustainable practices in modern education further shows the importance of aquaponics in the classroom.